PROJECT REPORT

AI-Based Student Performance Predictor With Personalized Intervention Suggestions

Course : Fundamental of AI/ML

Course code: CSA2001

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Technologies Used :

- Python
- Machine Learning (Linear Regression, Random Forest),
- Streamlit,
- Google Forms,
- Pandas
- Scikit-learn
- Matplotlib

1. Introduction

In today's learning system, student performance prediction is helpful in identifying underperforming students and improving academic performance. The conventional method only measures performance after exams, leaving little room for improvement.

The project proposes an AI system for predicting student performance using machine learning algorithms. The system provides personalized recommendations for improving performance. The system utilizes real-time data from students through Google Forms for accurate predictions.

2. Problem Statement

There are a number of students who perform poorly in their studies due to a number of reasons, including low attendance, poor learning habits, stress levels, and lack of consistency in learning, among others. However, there is no system that can effectively forecast the performance of the students and provide them with feedback.

The goal of the project:

- To forecast the performance of the students based on different academic and lifestyle-related variables
- To identify the students who are likely to perform poorly in their studies

3. Dataset Description

The data has been collected using Google Forms and consists of real data of the students.

Features:

- Attendance(%)
- Study hours/day
- Previous score
- Assignment completion(%)
- Sleep hours

- Screen time
- Study consistency(days/week)
- Stress level

Target Variable:

- Final score(out of 100)

4. Methodology

4.1 *Data Preprocessing*

- Removed unnecessary columns like 'Name', 'Timestamp'
- Renamed columns for consistency
- Handled missing values using mean imputation
- Converted data to a structured format

4.2 *Model Building*

Two different machine learning models are used:

- Linear Regression Model
- Random Forest Regression Model

Both models are built and evaluated; the best model is selected based on R^2 score.

4.3 *Model Evaluation*

The performance of the model is evaluated using:

- R^2 Score (Coefficient of Determination)

The model with a higher R^2 score is selected.

4.4 *Feature Importance*

For the Random Forest model, feature importance is calculated. Using this, it is determined which factors influence student performance in an important way.

Some example findings:

- Attendance is very important
- Study hours have an impact on performance
- Stress also affects performance

4.5 ***Prediction System***

The model predicts the final score based on user inputs through the application interface.

4.6 ***AI-Based Suggestion System***

A rule-based system is implemented for an AI-based suggestion system.

Some example findings:

- Poor attendance - Improve attendance
- Poor study hours - Improve study hours
- Poor stress - Improve stress

4.7 ***Study Plan Generation***

A study plan is generated based on the predicted score.

Some example findings:

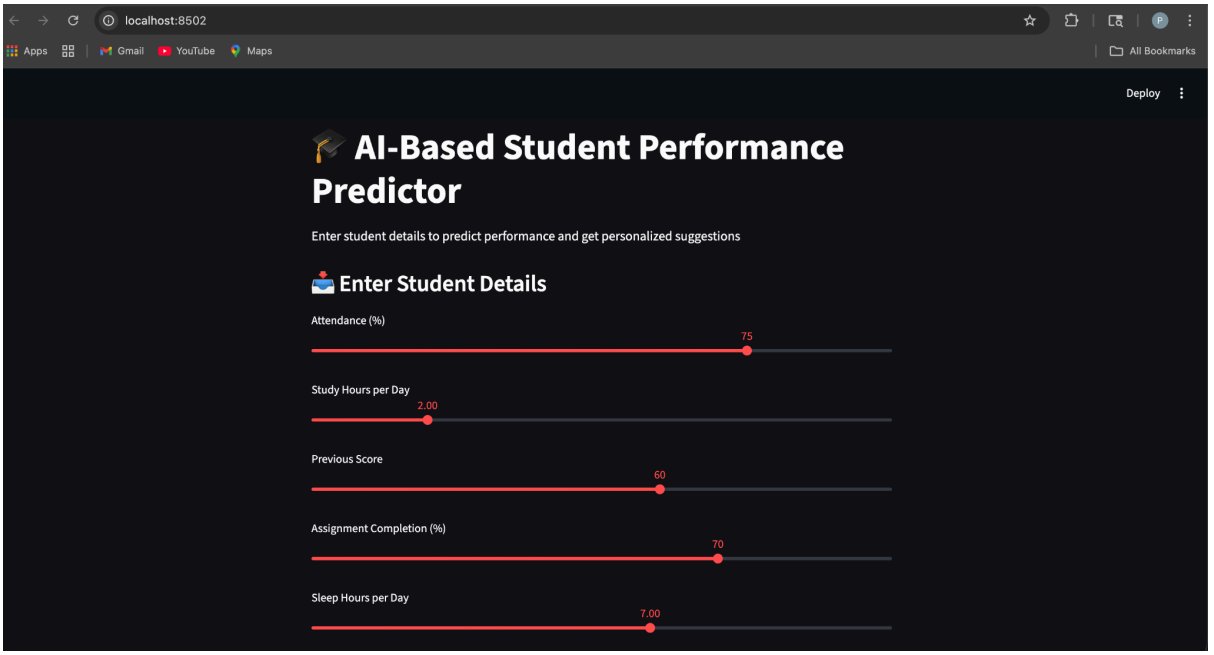
- Poor score - Study basics
- Medium score - Improve concepts
- High score - Move to advanced concepts

4.8 ***Web Application***

The system is implemented as an interactive web application using Streamlit.

Features:

- User input through sliders
- Real-time prediction
- Suggestion system
- Visualization graphs



5. Results

The system is able to predict student performance and give valuable insights.

The output of the system includes:

- Predicted score
- Grade
- Pass/Fail
- Risk level: High, Moderate, Low
- Personalized suggestions
- Weekly study plan
- Graphs of relationships:
- Study hours vs. performance
- Attendance vs. performance
- Stress vs. performance

Stress Level (1-10)

5

🔍 Predict Performance

📊 Prediction Result

Predicted Score: 88.62

Grade: A

Result: Pass

Risk Level: Low Risk 🟢

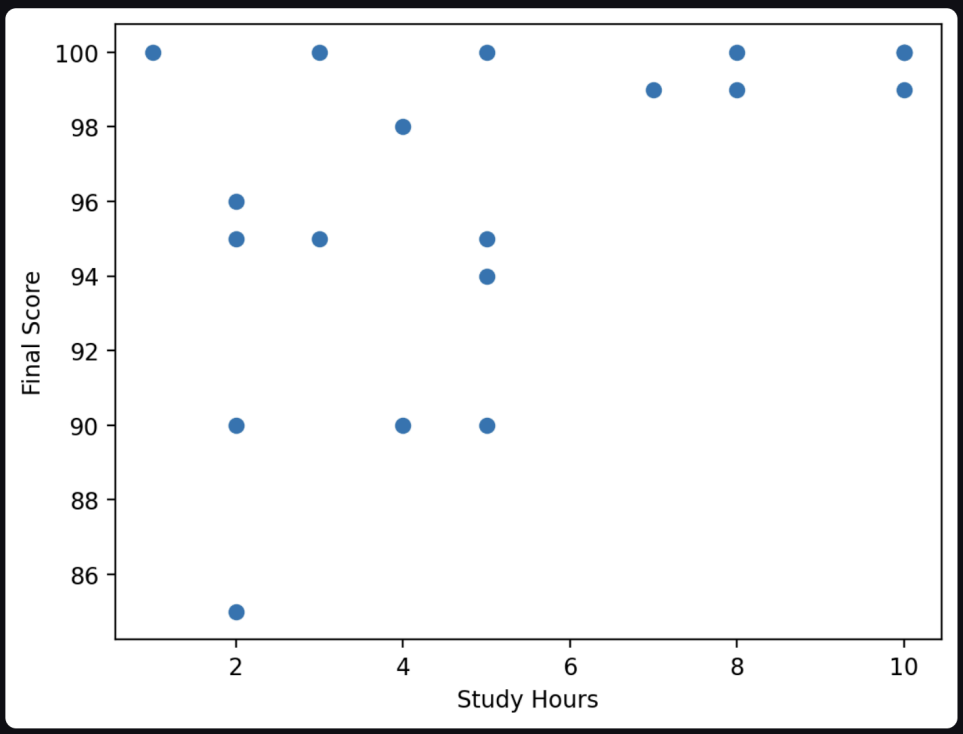
📌 Suggestions

- Try advanced topics and challenges

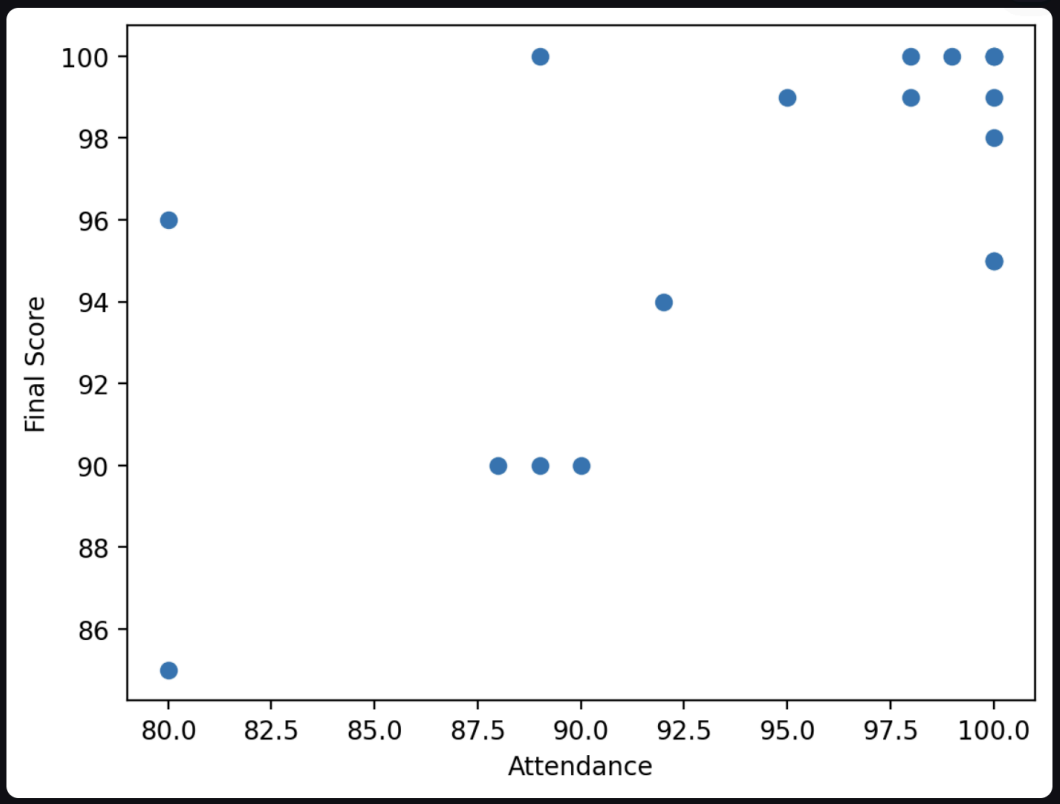
📅 Study Plan

- Day 1-3: Advanced topics
- Day 4-6: Projects/practice
- Day 7: Revision

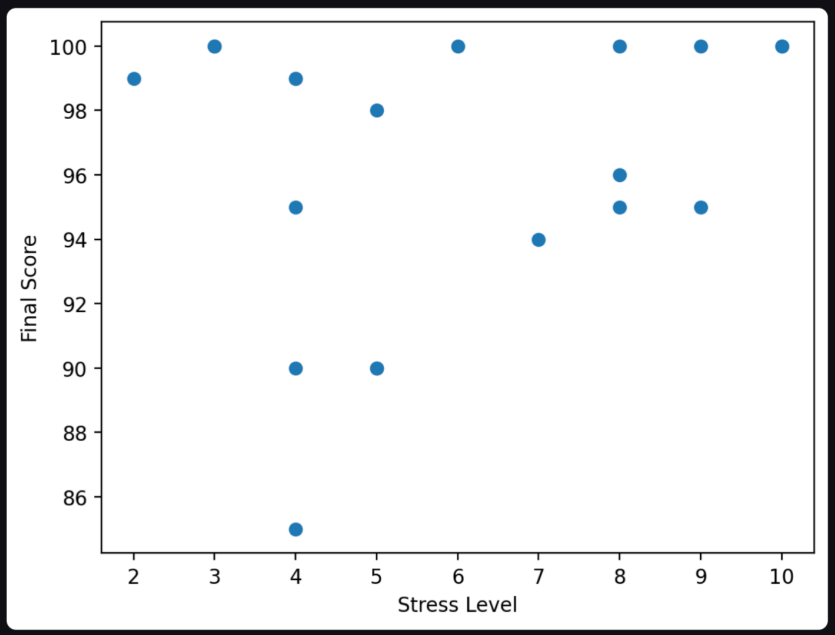
Study Hours vs Score



Attendance vs Score



Stress vs Score



6. Conclusion

In this project, we learned how machine learning can be used for the betterment of students by predicting performance and giving suggestions in a timely manner.

The system is very effective, user-friendly, and can be used in real-time for academic purposes.

7. Future Scope

- Add more advanced models (e.g., XGBoost)
- Use larger datasets
- Deploy as a mobile application
- Integrate real-time academic tracking

8. Project Executability

The project is fully executable from the command line without requiring any GUI-based development environment.

Steps to Run the Project:

1. Install dependencies:

```
pip install -r requirements.txt
```

2. Train the model:

```
python3 train_model.py
```

3. Run the application:

```
python3 -m streamlit run app.py
```

The application runs locally and opens in a web browser, allowing users to interact with the system.